Decision Trees, Ensemble Methods, and Random Forests

ML & Deep Learning Course

Pedro Watuhã dos Santos Guerra

April 3, 2026

Outline

1. Decision Trees
2. Ensemble Methods
3. Random Forests
4. Variable Importance
5. Benign Overfitting & Double Descent

[aspectratio=169]beamer []beamerthememetropolis []beamercolorthemedarkconsole tikz amsmath,amssymb

Decision Trees: Definition & Model

Intuition

Partition the feature space $\mathcal{X} \subseteq \mathbb{R}^p$ into *rectangular* regions $R_1, \ldots, R_M$ via recursive binary splits on a single feature at a time. Predict a constant in each region.

Mathematical Model

$$\hat{f}(\mathbf{x}) = \sum_{m=1}^{M} \hat{c}_m \, \mathbf{1}[\mathbf{x} \in R_m]$$

Parameter Estimation

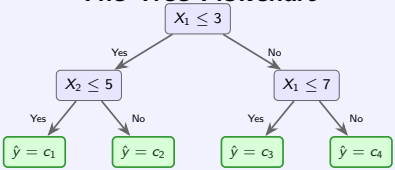
$$\hat{c}_m = \frac{1}{|R_m|} \sum_{\mathbf{x}_i \in R_m} y_i$$

(Regional average of training labels)

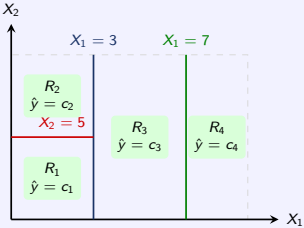
Decision Trees: Geometry & Properties

Geometric Interpretation: Recursive Splitting

The Tree Flowchart



Feature Space Partition



Key properties

- **Nonparametric:** Handles mixed feature types with ease.
- **Interpretable:** Direct map to a flowchart for easy decision-making.
- **High variance:** Small data perturbations yield drastically different trees.

Building the Tree: Recursive Binary Splitting

The Splitting Criterion (Regression)

At any given node, we search over all features j and all possible cutpoints s to define two half-planes:

$$R_1(j,s) = \{\mathbf{x} \mid X_j \leq s\} \quad \text{and} \quad R_2(j,s) = \{\mathbf{x} \mid X_j > s\}$$

We select the j and s that minimize the sum of squared residuals (RSS):

$$\min_{j,s} \left[\underbrace{\sum_{\mathbf{x}_i \in R_1(j,s)} (y_i - \hat{c}_1)^2}_{\text{Left region error}} + \underbrace{\sum_{\mathbf{x}_i \in R_2(j,s)} (y_i - \hat{c}_2)^2}_{\text{Right region error}} \right]$$

The Overfitting Trap

If left unconstrained, this recursive process will continue until every single data point is in its own rectangle (Train Error $= 0.00$).

Impurity Measures

At each internal node m with samples $\{y_i : x_i \in R_m\}$, measure the *impurity* to decide the best split.

Regression: Residual Sum of Squares

$$I(R_m) = \sum_{x_i \in R_m} (y_i - \bar{y}_{R_m})^2$$

Split reduces total RSS across two child nodes.

Classification: Cross-Entropy / Deviance

$$H(R_m) = -\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk}$$

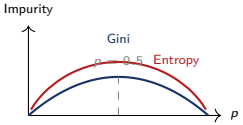
Also zero for pure nodes; convex in $\hat{p}$.

Classification: Gini Index

Let $\hat{p}_{mk}$ = fraction of class k in R_m:

$$G(R_m) = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk})$$

Zero when node is pure ($\hat{p}_{mk} = 1$ for some k).



Growing a Tree: The CART Algorithm

CART (Regression) Pseudocode

```
function GrowTree(R)
    if stopping criteria met then
        return leaf node with ĉ = (1/|R|) Σ_{x_i ∈ R} y_i
    end if
    // Find best feature j and split point s
    (j*, s*) ← arg min_{j,s} [ Σ_{x_i ∈ R_L} (y_i − ĉ_L)^2 + Σ_{x_i ∈ R_R} (y_i − ĉ_R)^2 ]
    Split region R into R_L, R_R at (j*, s*)
    return Node(j*, s*, GrowTree(R_L), GrowTree(R_R))
end function
```

The Intuition: Greedy Search

CART is **top-down and greedy**. It never looks ahead to see if a sub-optimal split now leads to a better tree later. It makes the locally optimal cut to minimize RSS *in the current moment*.

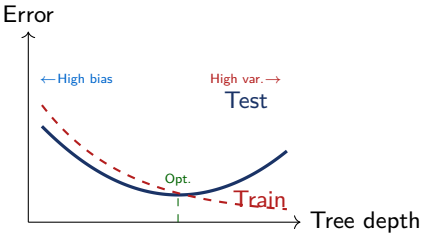
Bias–Variance Tradeoff in Trees

Deep trees:

- Low bias and High Variance
- Overfits training data

Shallow trees (large depth limit):

- High bias and Low variance
- Underfits complex structure



Bias–Variance decomposition

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{\text{Bias}^2[\hat{f}(x)]}_{\text{shallow tree}} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{deep tree}} + \sigma^2$$

Depth is the complexity parameter controlling this tradeoff.

Optimal depth selected by cross-validation or cost-complexity pruning.

Cost-Complexity Pruning

Motivation

A fully grown tree $T_{\max}$ overfits. We want the *smallest* subtree achieving near-optimal prediction: prune back from leaves, penalizing tree size.

Cost-Complexity Criterion

For a subtree $T \subseteq T_{\max}$ with $|T|$ leaves:

$$C_\alpha(T) = \underbrace{\sum_{m=1}^{|T|} \sum_{x_i \in R_m} (y_i - \hat{y}_{R_m})^2}_{\text{training RSS}} + \alpha |T|$$

$\alpha \geq 0$: complexity penalty (like Lasso's λ).

Weakest Link Pruning (Breiman 1984)

1. Start with full tree $T_{\max}$
2. Iteratively collapse the internal node t minimizing $\frac{\Delta \text{RSS}(t)}{|T_t|-1}$ (weakest link)
3. Produces nested sequence $T_{\max} \supset T_1 \supset \cdots \supset T_0$
4. Select α (hence T) by cross-validation

Strengths and Weaknesses of Single Trees

Strengths

- **Interpretable**: visual flowchart
- **No scaling required**: invariant to monotone feature transforms
- **Missing values**: surrogate splits
- **Non-parametric**: no distributional assumption

Weaknesses

- **High variance**: trees change drastically with small data changes
- **Axis-aligned boundaries**: poor for diagonal or smooth decision boundaries
- **Greedy splits**: not globally optimal partition
- **No uncertainty**: point prediction only

Minimax rate for trees

For a Lipschitz $f^\star$ in $\mathbb{R}^d$, the optimal tree estimator achieves:

$$\mathbb{E}\left[\left\|\hat{f}_n - f^\star\right\|^2\right] = O\left(n^{-2/(d+2)}\right)$$

Ensemble Methods: Motivation

Bias–Variance Decomposition (revisited)

For any estimator $\hat{f}$:

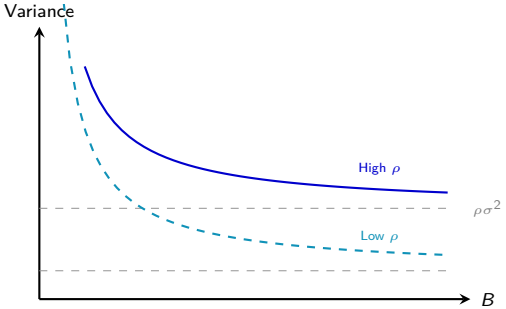
$$\mathbb{E}\big[(y - \hat{f}(x))^2\big] = \underbrace{\big(f^\star(x) - \mathbb{E}[\hat{f}(x)]\big)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}\big[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2\big]}_{\text{Variance}} + \sigma^2$$

Averaging reduces variance

Let $\hat{f}_1, \ldots, \hat{f}_B$ be identically distributed estimators with pairwise correlation ρ and individual variance σ^2. Then:

$$\mathrm{Var}\left(\frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)\right) = \rho\,\sigma^2 + \frac{1-\rho}{B}\,\sigma^2$$

As $B \to \infty$: residual variance $= \rho\,\sigma^2$.



Bagging vs Boosting: Overview

Bagging (Bootstrap Aggregating)

1. Draw B bootstrap samples $\mathcal{D}_1^*, \ldots, \mathcal{D}_B^*$ from $\mathcal{D}$
2. Fit base learner $\hat{f}_b$ on $\mathcal{D}_b^*$
3. Aggregate: $\hat{f}^{\text{bag}}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)$

Targets variance reduction.
Works best with high-variance, low-bias base learners (deep trees).

Boosting

1. Initialize $\hat{f}^{(0)}(x) = 0$
2. For $b = 1, \ldots, B$: fit $\hat{f}_b$ to *residuals* $r_i^{(b-1)} = y_i - \hat{f}^{(b-1)}(x_i)$
3. Update: $\hat{f}^{(b)} = \hat{f}^{(b-1)} + \nu\, \hat{f}_b \quad (\nu = \text{learning rate})$

Targets bias reduction.
Works best with high-bias, low-variance base learners (shallow trees).

Random Forests: Algorithm

Breiman's Random Forest (2001)

Random Forest = Bagging *with an additional randomization*: at each split, restrict the feature search to a random subset of $m \leq p$ features.

RF Algorithm

for $b = 1$ to B **do**
  Sample bootstrap $\mathcal{D}_b^* \sim \mathcal{D}$ (with replacement)
  Grow tree T_b on $\mathcal{D}_b^*$:
  while node not terminal **do**
    Sample m features uniformly at random
    Find best split among those m features
    Split node; descend
  end while
end for
Predict: $\hat{f}^{\mathrm{RF}}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$

Why feature subsampling?

Feature subsampling **forces different splits** at each tree, decorrelating predictions.

Out-of-Bag (OOB) error

Each bootstrap omits $\approx 36.8\%$ of data.
Evaluate each x_i on trees where $i \notin \mathcal{D}_b^*$:

$$\hat{\mathrm{Err}}^{\mathrm{OOB}} = \frac{1}{n}\sum_{i=1}^{n} L\left(y_i, \frac{1}{|\mathcal{B}_i|}\sum_{b \in \mathcal{B}_i} T_b(x_i)\right)$$

Consistency and Convergence of Random Forests

Theorem 1 (Breiman 2001 — informal)

The generalization error of a Random Forest converges a.s. as $B \to \infty$:

$$\mathrm{PE}^* \leq \frac{\bar{\rho}(1-s^2)}{s^2}$$

where $s =$ "strength" of individual trees and $\bar{\rho}$ is the mean correlation between residuals of any two trees.

Scornet, Biau & Vert (2015)

For *CART-based* random forests on additive models $f^\star(x) = \sum_{j=1}^{p} f_j(x_j)$:

$$\mathbb{E}\big[(\hat{f}_n(x) - f^\star(x))^2\big] \xrightarrow{n\to\infty} 0$$

First consistency result for the *actual* CART algorithm.

Biau (2012) — Centered forests

For a simplified "centered" RF with k_n leaves ($k_n \to \infty$, $k_n/n \to 0$):

$$\mathbb{E}\big[(\hat{f}_n(x) - f^\star(x))^2\big] \to 0 \quad \text{as } n \to \infty$$

Caveat: Rates

The exact minimax rate of RF is still open in full generality. For Lipschitz $f^\star$ in $\mathbb{R}^d$, the best known adaptable rate is

$$\tilde{O}\left(n^{-\frac{2}{d+2}}\right)$$

Convergence Rate: Adaptivity to Sparsity

The Curse of Dimensionality (Recall)

For a Lipschitz continuous function $f^\star$ in $\mathbb{R}^p$, standard non-parametric estimators achieve a rate of $O(n^{-2/(p+2)})$. If the number of features p is large, the rate stalls and learning stops.

Theorem 2 ((Rate of Convergence for Centered RF — Biau, 2012))

Assume $f^\star$ depends only on a small subset of S active features out of p total features ($S \ll p$). For a centered Random Forest $\hat{f}_{RF}$, the mean squared error converges at a rate of:

$$\mathbb{E}\left[(\hat{f}_{RF}(\mathbf{x}) - f^\star(\mathbf{x}))^2\right] = O\left(n^{\frac{-3}{3+4S\log 2}}\right)$$

Why this is a breakthrough result

- **Bypassing p:** The exponent depends *only* on the intrinsic dimension S, completely bypassing the physical dimension p.

Why RF Outperforms Single Trees in Practice

Variance reduction

Individual tree variance σ^2, correlation ρ:

$$\mathrm{Var}(\hat{f}^{\mathrm{RF}}) = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B}$$

RF achieves low ρ via feature subsampling. As $B \to \infty$, floor is $\rho\sigma^2$ which is small when $m \ll p$.

Bias

Averaging unbiased estimators preserves unbiasedness:

$$\mathrm{Bias}[\hat{f}^{\mathrm{RF}}] = \mathrm{Bias}[\hat{f}_b]$$

Each fully grown tree has (approximately) zero bias.

Variable Importance: MDI

Mean Decrease in Impurity (MDI) / Gini Importance

For feature j, sum the weighted impurity reductions over all splits on X_j, averaged over trees:

$$\widehat{\mathrm{MDI}}(j) = \frac{1}{B}\sum_{b=1}^{B} \sum_{\substack{v \in T_b \\ \text{split on } X_j}} p_v \cdot \Delta I(v)$$

where $p_v = n_v/n$ is the fraction of samples reaching node v, and $\Delta I(v) = I(\text{parent}) - \frac{n_L}{n_v} I(L) - \frac{n_R}{n_v} I(R)$.

Advantages

- Computed for free during training
- Fast: $O(1)$ lookup after fitting
- Works for any impurity measure (Gini, entropy, RSS)

Known biases (Strobl et al. 2007)

- **Cardinality bias**: continuous or high-cardinality features get inflated MDI even if irrelevant
- **Correlated features**: importance is split arbitrarily among correlated variables

Variable Importance: MDA (Permutation Importance)

Mean Decrease in Accuracy (Breiman 2001)

For each tree T_b, evaluate OOB error, then permute feature j in the OOB set and re-evaluate:

$$\widehat{\mathrm{MDA}}(j) = \frac{1}{B}\sum_{b=1}^{B}\Big[\underbrace{\mathrm{Err}_{\pi_j}^{(b)}}_{\text{permuted}} - \underbrace{\mathrm{Err}_{\mathrm{OOB}}^{(b)}}_{\text{original}}\Big]$$

Permuting X_j breaks its association with Y — if X_j is important, error increases.

Advantages

- **Model-agnostic**: works post-fit for any model
- **Less biased** by cardinality than MDI
- Can be calibrated: compare to random noise features

Limitations

- **Correlated features**: permuting X_j creates unrealistic feature combinations; still splits credit arbitrarily
- **U**nstable and Slower.

Variable Importance: Failure Modes

Correlated features problem

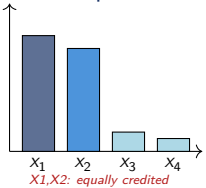
Suppose $X_1 \approx X_2$ (e.g. $\rho = 0.95$) and only X_1 truly predicts Y.

- **MDI**: both X_1 and X_2 are used interchangeably in splits $\Rightarrow$ importance split roughly 50/50 (misleading)
- **MDA**: permuting X_1 hurts less because X_2 compensates $\Rightarrow$ underestimates X_1; permuting X_2 also hurts (spuriously high)

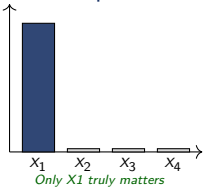
Recommendation

Use **SHAP values** (Lundberg & Lee 2017) or **conditional permutation** (Strobl et al. 2008) for more reliable importance in correlated settings.

MDI Importance



True importance



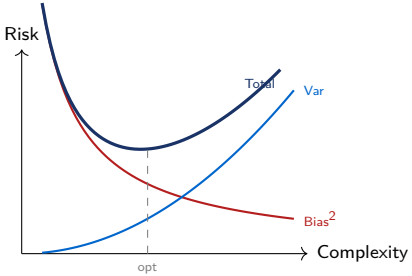
Classical Wisdom: The Bias–Variance U-Curve

Reminder from Set 1

For any estimator $\hat{f}$ trained on n samples:

$$\underbrace{L(\hat{f}) - \inf_{f} L(f)}_{\text{excess risk}} \leq \underbrace{\inf_{f \in \mathcal{H}} L(f)}_{\text{approx. error}} + \underbrace{2\mathcal{R}_n(\mathcal{H})}_{\text{est. error}}$$

- Larger $\mathcal{H} \Rightarrow$ lower bias, higher Rademacher complexity
- Classical recipe: tune capacity to minimise the sum
- Result: a U-shaped risk curve in model complexity



Classical ERM generalisation bound (Set 1):

$$L(\hat{f}_{\text{erm}}) \lesssim \inf_{f \in \mathcal{H}} L(f) + \frac{C}{\sqrt{n}}$$

Classical principle

Fitting training data *perfectly* (zero empirical risk) with a rich model class should lead to poor generalisation.

The Interpolation Regime

What "interpolating" means

A predictor $\hat{f}$ is **interpolating** if it fits all training labels exactly:

$$\hat{f}(\mathbf{x}_i) = y_i, \quad i = 1, \ldots, n.$$

Equivalently, *empirical risk is zero*: $\widehat{L}(\hat{f}) = 0$.

Why classical theory predicts failure

- Uniform convergence bounds: $\sup_{f \in \mathcal{H}} |L(f) - \widehat{L}(f)|$ must become vacuous once $\mathcal{H}$ is rich enough to interpolate noisy data (excess risk $\geq$ noise level)
- VC/Rademacher bounds diverge as $|\mathcal{H}| \to \infty$
- Classical theory says: *control complexity or suffer*

Empirical surprise

- Overparameterised neural networks ($d \gg n$) interpolate noisy training data
- Yet test accuracy remains excellent
- No explicit regularisation needed

Key question: Can interpolation ever be *benign*? Under what conditions does fitting noise not hurt?

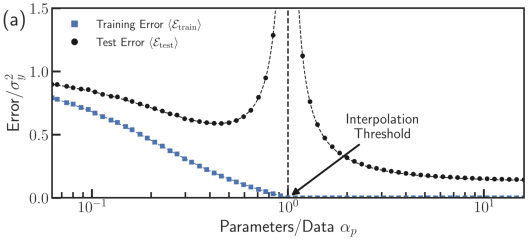
Double Descent

Beyond the classical U-curve

As model capacity increases past the *interpolation threshold* ($d \approx n$), the risk curve exhibits a **second descent**:

1. **Classical regime** ($d \ll n$): standard bias-variance U-curve
2. **Interpolation threshold** ($d \approx n$): risk peaks (model is just barely able to fit the data — poor generalisation)
3. **Modern regime** ($d \gg n$): risk *decreases* again as overparameterisation grows

The second descent was observed empirically (Belkin et al. 2019) and linked to minimum-norm interpolation.



Intuition

More parameters $\Rightarrow$ more solutions that interpolate; optimisation selects the *simplest* one (minimum norm), which can generalise well.

Effective Rank: When Is Overfitting Benign?

The key quantity: effective rank

For a partition index k, define the **effective rank** of the tail:

$$r_k(\mathbf{\Sigma}) \;:=\; \frac{\sum_{i>k} \lambda_i}{\lambda_{k+1}}.$$

- $r_k(\mathbf{\Sigma}) \geq bn$ means the tail eigenvalues are *near-isotropic at scale* λ_{k+1}
- In this regime $X_{>k} X_{>k}^\top \approx \gamma\, \mathbf{I}_n$: self-induced regularisation of order $\gamma = \sum_{i>k} \lambda_i$

Benign overfitting theorem (informal)

Benign overfitting for the minimum-norm interpolant requires:

1. **Simple signal**: the prediction component lives in a *low-dimensional* ($k \ll n$) high-variance subspace $\Rightarrow$ small bias and k/n variance
2. **Rich tail**: effective rank $r_k(\mathbf{\Sigma}) \gg n \Rightarrow$ overfitting component variance is small

Connection to Random Forests

Random Forests as overparameterised ensembles

- Each tree is a **high-variance interpolating estimator** (deep trees memorise training data exactly)
- Averaging B trees: variance $\to \rho\sigma^2$ (residual controlled by inter-tree correlation ρ)
- **Feature subsampling** ($m < p$ features per split) actively *diversifies* the spectral directions each tree uses
- Analogous to sampling from different eigenspaces: the ensemble explores the full feature covariance

Benign overfitting lens

- Individual trees: *harmful* overfitting (high variance)
- Forest average: effective rank of the feature covariance governs whether the aggregate variance is controlled
- Feature subsampling $\Rightarrow$ decorrelation $\Rightarrow$ implicit diversity in eigenspace coverage
- High-dimensional settings ($p \gg n$) benefit most: more room for the tail eigenvalues to provide near-isotropic averaging

Intuition: RF manufactures the spectral diversity that makes benign overfitting possible in the aggregate.